PRESIDENT OFFICE,

REGIONAL ADMINSTRATION AND LOCAL GOVERNMENT

NAMANGA SECONDARY SCHOOL

FORM THREE PHYSICS MID-TERM TEST

TIME: 3HOURS **APRIL 2025**

Instructions

1 This paper consists of eleven (11) questions. With section A, B and C

2 Answer all questions in section A, B and two questions in section C

3 all written must be in blue or black ink and drawings in pencils

4 the following constant may be used

Density of water =1000Kg/m^3 acceleration due to gravity (g) =9.8m/s^2

SECTION A (16 marks)

Answer **all** questions in this section

1. For each of the items (i) – (x), choose the correct answer among the given alternatives and write its letter beside the item number in the answer sheets provided.

(i) The irregular motion of tiny particles suspended in a fluid is called;

A. Mobility B. Kinetic motion C. Random motion
D.Brownian motion. E. Molecular motion

(ii) The area under velocity-time graph represents.

A.Acceleration B. Force C. Distance
D Momentum E. Speed

(iii) The property of the material to recover its original shape and size on removal of a stretching force is called

A. Elasticity B. Plasticity C. Hooke's law
D. Cohesively E. Adhesive force

(iv) A vector has both magnitude and direction and the examples are;

A. Work and displacement B. Displacement and speed.
C. Speed and momentum D. Acceleration and momentum.
E. Energy and pressure

(v). The laws of vector addition are;

A. The triangle law only B.The triangle law and the polygon law only

C. The polygon law, triangle law and the parallelogram law

D. The parallelogram law.

E. Sine rule, cosine rule and Pythagoras theorem

(vi). The forces of 4N and 3N act on a body due East and North respectively. The resultant force is;

A. 5N, 36°N.E B. 5N, 45°N.E C. 5N, S.W

D. 5N, 45°N.E E. 7N, 450 N.E

(vii). The relative velocity of a body is the velocity of a body with respect to another;

A. Body at rest B. Moving body or at rest

B. Body of smaller mass D. Body of higher velocity.

E. Moving body

(viii). A floating body experiences an upthrust which is equal in magnitude to the;

A. Weight of the fluid displaced

B. Weight of the fluid in which the body floats.

C. Density of the fluid displaced

D. Weight of the body when filled with air

E. Mass of fluid displaced

(ix). A bus carrying a very big load on its carrier can easily overturn because;

A. It cannot run fast B. Its equilibrium is stable

C. Its Centre of gravity is low D. Its Centre of gravity is high

E. Its Centre of gravity is at it's Centre

(x) Which of the following statements is true about the image formed by curved mirrors?

A.All real images are reduced B.All virtual images are inverted

C. All inverted images are enlarged D. All real images are inverted

E. All virtual images are enlarged

2. Match the items in **List A** with responses in **List B** by writing the letter of the correct response beside the item number in the answer booklet provided.

List A	List B
i. Umbra	A. Tinted glass
ii. Real image	B. Glass block
iii. Penumbra	C. Pin hole camera
iv. Virtual image	D. Plane minor
v. Transparent	E. Regular reflection
vi. Translucent	F. Irregular reflection
	G. Partial shadow
	H. Total shadow

SECTION B (54 Marks)

Answer **all** questions in this section

3. (a)What effect does an increase in temperature have on the density of most liquids?
   (b) State Archimedes principle.
   (c) When a piece of wood is put in graduated cylinder containing water, the level of the water rises from 17.7cm^3 to 18.5 cm^3. Calculate the
   (i)Mass of a piece of wood
   (ii) Total volume of a piece of wood given that its relative density is 0.60.

4. (a)Define the term friction.
   (b) State the laws of friction.
   (c) A box of mass 50kg resting on a wooden floor. Calculate the frictional force if the coefficient of static friction is 0.5
5. (a) State laws of refraction of light
   (b) A ray of light at an angle of 60 on a block of glass of refractive index 1.5. Determine the angle of refraction of the ray
   (c) A rectangular glass block 5cm thick is placed on the top of the page of a book. If the refractive index of the glass block is 1.53. Calculate apparent depth of the letters on the book.

6. (a)State and describe two factors on which pressure exerted by the liquid depends.
   (b) Hydraulic press has a large circular piston of diameter 0.7m and circular piston to which the effort is applied is 0.2m. A force of 300N is exerted on a small piston. Find the force required to lift the heavy load.
7. (a) Whenever you want to remove a very tight nut from your bicycle you have to choose a longer spanner, why is it so?

(b) A rocket expels gas at the rate of 0.5kg/s. If the force produced by the rocket is 100 N, what is the velocity with which the gas is expelled?

(c) Describe the resolution of a vector V making an angle β with the horizontal in X-Y plane.

8. (a)By giving three examples in each one differentiate between a scalar quantity and a vector quantity.

   (b) (i) State parallelogram law of vector addition

   (ii) Use the law you have stated in (b)(i) to find the resultant of two forces of magnitudes 50N and 70N acting at a point making an angle of 45^0 between them.

SECTION C (30 marks)

Answer only **two (2)** questions in this section

9. A vehicle moves from rest with a uniform acceleration of 1.5m/s^2 for the first 10s and continues accelerating at 0.5m/$s^2$for a further 20s. it continues at constant speed for 90s and finally takes 30s to decelerate uniformly to rest.

   (a) Draw a graph of speed against time for the journey.

   (b) From your graph, deduce the total distance travelled.

   (c) What is the average speed of the train for the whole journey?

10. (a) Distinguish between absolute velocity and relative velocity.

    (b) The relative velocity of car A with respect to car B is 5m/s when they are moving in the same direction and 25m/s when they are moving in opposite direction; find the velocity of each car.

    (c) A trolley of weight 200N is being pushed by a force of 100N making an angle 30^0 with the horizontal. Calculate the horizontal and vertical components of the force and hence the total downward force.

11. (a) Mention three advantages of friction and three disadvantages of friction.

    (b) Mention three ways of increasing friction and three ways of reducing friction.

    (c) Explain how friction occurs.